



Project Executable Files

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
/credit-card-approval-prediction
```

```
|
```

```
├── /backend                                # FastAPI backend server
│   ├── /api                              # Route definitions and endpoints
│   ├── /models                           # Saved XGBoost ML model and scalers
│   ├── /services                         # ML prediction and SHAP calculation logic
│   ├── main.py                           # Entry point for backend
│   └── requirements.txt                   # Python dependencies
├── /frontend                              # React agent dashboard
│   ├── /src                              # Source code (components, API calls)
│   ├── /public                           # Static assets
│   └── package.json                       # Node.js dependencies
├── /database                             # PostgreSQL schema & migration scripts
├── .env.example                           # Template for environment variables
├── docker-compose.yml                     # Docker config to run both services
└── README.md                             # General setup and documentation
```

Step 3: Deployment / Access Details

Hosted / Deployed URL	https://credit-card-predictor-demo.vercel.app
Repository Link	https://github.com/xxxxxx/credit-card-approval-prediction
Hosting Provider / Platform	AWS EC2 (Backend API), Vercel (Frontend UI)

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- Python 3.9+ installed
- Node.js (v16+) installed
- PostgreSQL installed and running on port 5432

Step 1: Clone the repository:

```
git clone https://github.com/xxxxxx/credit-card-approval-prediction.git
cd credit-card-approval-prediction
```

Step 2: Configure Environment:

Copy ``.env.example`` to ``.env`` in both `/backend` and `/frontend` directories.
Update the `POSTGRES_URL` in the backend ``.env`` file.

Step 3: Setup Backend (FastAPI):

```
cd backend
```

```
pip install -r requirements.txt
uvicorn main:app --reload --port 8000
```

Step 4: Setup Frontend (React):

```
cd ../frontend
npm install
npm start
```

Step 5: Access the application:

Open your browser and navigate to <http://localhost:3000>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Initial startup of the SHAP explainer module can take ~10 seconds.	The explainer is now cached in memory after the first request. Status: Mitigated.
2	Frontend UI is not fully optimized for mobile screens smaller than 375px.	Agents are expected to use desktop/tablets. Status: Pending media query updates.
3	No automated retraining pipeline for data drift yet.	Admins must manually trigger the training script via CLI. Status: Planned for V2.